AI-Invigilator

MULTI-MODAL AUTOMATED EXAM PROCTORING SYSTEM

COMPREHENSIVE PROJECT OVERVIEW & VIVA PRESENTATION GUIDE

1. Executive Synopsis

AI-Invigilator is an advanced, full-stack online examination proctoring system that transforms standard computer webcams and microphones into a Multi-Modal AI Security Suite (Vision + Audio). It addresses the core vulnerabilities of remote testing—impersonation, off-camera whispering, mobile phone usage, and tab switching—without requiring manual invigilation for every candidate.

2. Core System Modules

A. Pre-Exam Identity AI (Anti-Impersonation)

Captures a baseline face photo during registration and performs 3D landmark vector matching before unlocking exam questions.

B. Vision AI Monitoring Engine

Uses MediaPipe 3D FaceLandmarker (468 points for eye tracking and head direction) and TensorFlow.js COCO-SSD for mobile phone & multi-person detection at 30 FPS.

C. Multi-Modal Audio & Whisper Isolator

Uses Web Audio API & FFT Spectrum Analysis to isolate human vocal formants (120Hz-2.4kHz) and off-camera whispering fricatives (2.5kHz-5.5kHz) from background fan noise.

D. Live Proctor Dashboard & Tap-In Console

Real-time traffic-light grid (Normal, Warning, Suspicious) with silent live stream access (webcam + screen share) and PDF report generation.

3. Technical Stack

LAYER	TECHNOLOGIES USED
Frontend UI	React 18, Vite, Tailwind CSS, Material Symbols
Client-Side AI/ML	MediaPipe Tasks Vision, TensorFlow.js COCO-SSD, Web Audio API
Backend & API	Node.js, Express.js, JWT Authentication
Database & Reports	PostgreSQL (ai_invigilator), jsPDF, autoTable

4. Team Member Topic Division & Viva Roles

MEMBER 1: Vision AI & Identity Verification

Focus: Pre-Exam Identity Verification, MediaPipe 3D Mesh, TensorFlow COCO-SSD Phone Detection.

Q: How does identity verification work?

Ans: Extracts facial landmarks during registration; matches live camera vectors using Euclidean distance.

Q: How do AI models run client-side?

Ans: Executed via WebGL hardware acceleration at 30 FPS without server latency.

MEMBER 2: Multi-Modal Audio AI & Active Exam Engine

Focus: Web Audio API, FFT Spectrum Analysis, Whisper vs Speech Isolator, Dynamic AI Trust Scoring.

Q: How is off-camera whispering detected?

Ans: Uses Biquad bandpass filter (120Hz-4.5kHz). Isolates 2.5kHz-5.5kHz sibilant fricative harmonics from fan noise.

Q: How is the dynamic AI Trust Score calculated?

Ans: Starts at 98%. Deducts -55% Phone, -45% Multi-Face, -35% Speech, -25% Whisper, -15% Head Turn.

MEMBER 3: Proctor Dashboard & PDF Analytics

Focus: Teacher Live Monitoring Grid, Silent Live Stream Tap-In, Postgres Schema, PDF Reports.

Q: How does the proctor monitor candidates in real-time?

Ans: Active exam pings telemetry every 2s; dashboard categorizes into Normal, Warning, and Suspicious.

Q: What does the View Report modal do?

Ans: Displays in-app PDF preview with student registration photo, score, risk score, and incident log.

5. 30-Second Opening Elevator Pitch

RECOMMENDED PRESENTATION OPENING:

"Good morning Madam. Today we present AI-Invigilator, a Multi-Modal AI automated online exam proctoring system. Traditional online exams suffer from impersonation, off-camera whispering, mobile phone usage, and tab switching. Our application solves this by combining client-side Computer Vision (MediaPipe and TensorFlow.js) with Web Audio API frequency spectrum analysis. We verify candidate identity before the exam, track vision and audio harmonics in real time, compute a dynamic AI Trust Score, and provide teachers with a Live Traffic-Light Monitoring Dashboard and automated PDF report cards containing student baseline photos."